From Luminal Kinematics to Informational Causality: Evaluating the Theoretical and Empirical Consequences of Replacing the Speed of Light with the Speed of Information

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Abstract

For over a century, the fundamental kinematic invariant of the universe, $c \approx 2.99792458 \times 10^8$ m/s, has carried the historical nomenclature "the speed of light in vacuum." While historically understandable given the derivation of Maxwell's electrodynamics, this terminology has fostered persistent pedagogical confusions, obscured the axiomatic structure of spacetime, and detached the kinematic foundation of relativity from modern developments in quantum information theory, many-body physics, and holographic quantum gravity. In this study, we conduct an extensive theoretical and computational investigation into whether replacing the concept of the "speed of light" (c) with the "speed of information" ($c_{\mathcal{I}}$) provides a deeper, more fundamental understanding of relativistic kinematics, quantum measurement, and emergent spacetime. We formalize this transition across four distinct tiers: (i) **Axiomatic Spacetime Foundations**, where we prove via group-theoretic and causal order methods that spacetime geometry requires only an invariant causal velocity $c_{\mathcal{I}}$ decoupled from any specific gauge field or massless particle; (ii) **Wave Dynamics & Dispersion Analysis**, demonstrating via Kramers-Kronig analyticity and numerical simulations that while phase and group velocities can exceed c or become negative in dispersive media, the informational front velocity is strictly and universally bounded by $c_{\mathcal{I}}$; (iii) **Microcausality and Quantum Information**, illustrating how the no-signaling theorem and Lieb-Robinson bounds in non-relativistic many-body Hamiltonians naturally generate emergent causal information cones ($v_{\mathrm{LR}} \leq c_{\mathcal{I}}$); and (iv) **Holographic Gravity & Planck-Scale Physics**, where butterfly velocities v_B, out-of-time-ordered correlators (OTOCs), and modified dispersion relations (MDRs) position information flow as the ontological precursor to macroscopic spacetime metric structures. We conclude that while a purely nominal substitution ($c \to c_{\mathcal{I}}$) leaves the quantitative predictions of the standard Einstein-Maxwell action invariant under the condition $m_\gamma = 0$, the conceptual and structural shift to $c_{\mathcal{I}}$ clarifies fundamental physics: it resolves long-standing interpretive paradoxes, correctly decouples geometry from gauge-field contingencies (such as non-zero photon mass in Proca theories), and provides the foundational vocabulary required for the "It-from-Qubit" emergence of spacetime in quantum gravity.

Keywords: Foundations of Relativity, Quantum Information, Microcausality, Lieb-Robinson Bounds, Holography, Modified Dispersion Relations.

1 Introduction: The Historical Accident of Maxwellian Relativism

Albert Einstein's 1905 paper, *Zur Elektrodynamik bewegter Körper* ("On the Electrodynamics of Moving Bodies"), rested on two fundamental postulates:

1. **The Principle of Relativity:** The laws of electrodynamics and optics are valid for all frames of reference for which the equations of mechanics hold good.

2. **The Constancy of the Speed of Light:** Light is always propagated in empty space with a definite velocity c which is independent of the state of motion of the emitting body.

From a historical perspective, Einstein's second postulate was an act of physical genius that reconciled the null result of the Michelson-Morley experiment with Maxwell's wave equation:

$$\nabla^2 \mathbf{E} - \frac{1}{c^2}\frac{\partial^2 \mathbf{E}}{\partial t^2} = 0, \quad \text{where } c = \frac{1}{\sqrt{\epsilon_0 \mu_0}}. \tag{1}$$

However, designating c as "the speed of light" embedded an incidental material phenomenon—the propagation velocity of transverse excitations in the $U(1)$ electromagnetic gauge field—into the very definition of spacetime geometry. This historical conflation has generated substantial conceptual, pedagogical, and physical difficulties:

1. **Pedagogical and Ontological Vulnerability:** When light slows down to 17 m/s in a Bose-Einstein condensate or to 2.25×10^8 m/s in water, students routinely ask if the fabric of spacetime or the relativistic limit has shifted.

2. **Contingency on Particle Physics:** If the photon should possess an infinitesimal rest mass $m_\gamma \neq 0$ (as permitted in de Broglie-Proca electrodynamics), electromagnetic waves would disperse and propagate at velocities strictly less than c. Yet, the geometry of Minkowski spacetime, time dilation, mass-energy equivalence $E = mc^2$, and causality would remain completely unaffected.

3. **Apparent Superluminal Paradoxes:** Experiments demonstrating phase velocities $v_p > c$ or group velocities $v_g > c$ (and negative group delays) in anomalous dispersion media routinely trigger specious claims regarding the breakdown of relativity.

4. **Disconnection from Modern Quantum Gravity:** In modern holographic dualities (AdS/CFT), tensor network geometries (MERA), and quantum information approaches to spacetime, geometry is recognized as an *emergent* phenomenon arising from quantum entanglement and information processing.

The central question of this study is: **Does replacing the concept of the "speed of light" with the "speed of information" ($c_{\mathcal{I}}$) help us better understand the fundamental architecture of spacetime, quantum causality, and gravitational dynamics?**

2 Axiomatic Spacetime Foundations: Spacetime Kinematics without Electromagnetism

2.1 The Ignatowsky-Frank-Rothe-Lévy-Leblond Theorem

It is a widespread misconception that Special Relativity fundamentally depends upon the existence of light. As early as 1910, W. von Ignatowsky demonstrated that the Lorentz transformations can be derived without the second postulate, relying purely on spatial isotropy, homogeneity of spacetime, and the group structure of inertial reference frames. This was refined by Frank and Rothe (1911), Lévy-Leblond (1976), and Pelissetto & Testa (2015).

Let an event in inertial frame K be (t, x) and in frame K' (moving with velocity v relative to K) be (t', x'). By spacetime homogeneity, the coordinate transformation must be linear:

$$\begin{pmatrix} x' \\ t' \end{pmatrix} = \begin{pmatrix} A(v) & B(v) \\ C(v) & D(v) \end{pmatrix} \begin{pmatrix} x \\ t \end{pmatrix}. \tag{2}$$

Applying the standard symmetries:

1. **Homogeneity and Isotropy of Space:** Reflections $x \to -x$ and spatial inversions preserve physical laws.
2. **Subgroup Closure and Reciprocity:** Two successive boosts in the same direction must yield a valid boost: $\mathcal{B}(v_2) \circ \mathcal{B}(v_1) = \mathcal{B}(v_3)$, with $\mathcal{B}(-v) = \mathcal{B}(v)^{-1}$.
3. **Causal Ordering:** The temporal sequence of causally connected events cannot invert under continuous frame transformations.

From these purely group-theoretic axioms, the transformation coefficients are uniquely constrained to:

$$x' = \gamma(v)(x - vt), \quad t' = \gamma(v)\,(t - \alpha v x)\,, \quad \gamma(v) = \frac{1}{\sqrt{1 - \alpha v^2}}, \tag{3}$$

where α is a universal invariant constant of nature with dimensions of $[\text{velocity}]^{-2}$.

There are precisely three mathematically permissible cases:

- **Case 1:** $\alpha < 0$**.** Spacetime symmetry group is $SO(4)$, which permits periodic time loops (closed timelike curves) in ordinary inertial transformations, violating causality.
- **Case 2:** $\alpha = 0$**.** Spacetime symmetry group is the Galilean group $\mathrm{SGal}(3,1)$, where the causal speed limit is infinite ($\alpha = 1/\infty^2 = 0$).
- **Case 3:** $\alpha > 0$**.** Spacetime symmetry group is the Poincaré / Lorentz group $SO(3,1)$, with an invariant universal speed:

$$c_{\mathcal{I}} \equiv \frac{1}{\sqrt{\alpha}}. \tag{4}$$

Notice that **not a single mention of light, electromagnetism, photons, or Maxwell's equations is required** to arrive at the invariant speed $c_{\mathcal{I}}$. The invariant speed is an intrinsic geometric feature of our pseudo-Riemannian manifold: it represents the maximal causal speed at which two events can exchange state alterations—i.e., the **speed of information**.

2.2 Decoupling Field Excitations: Maxwell vs. Proca Electrodynamics

Why did physics conflate $c_{\mathcal{I}}$ with light? Because in standard Maxwellian electrodynamics, the photon is assumed to be strictly massless ($m_\gamma = 0$):

$$\mathcal{L}_{\text{Maxwell}} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu}. \tag{5}$$

In Lorenz gauge $\partial_\mu A^\mu = 0$, the wave equation reads:

$$\Box A^\mu \equiv \left(-\frac{1}{c_{\mathcal{I}}^2} \frac{\partial^2}{\partial t^2} + \nabla^2 \right) A^\mu = 0 \implies \omega^2 - c_{\mathcal{I}}^2 k^2 = 0 \implies v_{\text{group}} = c_{\mathcal{I}}. \tag{6}$$

However, in de Broglie-Proca electrodynamics, where the photon carries rest mass m_γ:

$$\mathcal{L}_{\text{Proca}} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + \frac{1}{2} \left(\frac{m_\gamma c_{\mathcal{I}}}{\hbar} \right)^2 A_\mu A^\mu. \tag{7}$$

The dispersion relation becomes:

$$\omega(k) = \sqrt{c_{\mathcal{I}}^2 k^2 + \frac{m_\gamma^2 c_{\mathcal{I}}^4}{\hbar^2}}, \tag{8}$$

yielding a group velocity of light $v_\gamma(k)$:

$$v_{\text{group}} = \frac{d\omega}{dk} = c_{\mathcal{I}} \sqrt{1 - \frac{m_\gamma^2 c_{\mathcal{I}}^4}{\hbar^2 \omega^2}} < c_{\mathcal{I}}. \tag{9}$$

Under Proca electrodynamics:

1. Low-frequency radio waves travel significantly slower than high-energy gamma rays.
2. Near the cutoff frequency $\omega_{\text{cutoff}} = m_\gamma c_{\mathcal{I}}^2/\hbar$, the group velocity approaches zero.
3. Yet, throughout this scenario, $c_{\mathcal{I}}$ **remains entirely invariant, constant, and unyielding**.

Thus:

$$\boxed{c_{\mathcal{I}} \text{ is the invariant metric asymptote; } v_\gamma(\omega) \text{ is a dynamical field excitation velocity.}} \tag{10}$$

3 Wave Dynamics & Dispersion: Optical Artifacts vs. The Informational Front

3.1 Sommerfeld-Brillouin Contour Analysis and Kramers-Kronig Analyticity

To prove that no physical information can propagate faster than $c_{\mathcal{I}}$ regardless of the medium's refractive index $n(\omega)$, we examine the propagation of a scalar field $E(x,t)$ through a dispersive medium:

$$E(x,t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \tilde{E}(0,\omega)\exp\left[i\left(\frac{\omega n(\omega)}{c_{\mathcal{I}}}x - \omega t\right)\right]d\omega. \tag{11}$$

Suppose a transmitter switches on an arbitrary signal at $t = 0$: $E(0,t) = 0$ for all $t < 0$. By Titchmarsh's theorem, the Fourier transform $\tilde{E}(0,\omega)$ is analytic in the upper half of the complex frequency plane $\mathbb{C}^+ = \{\omega \in \mathbb{C} \mid \text{Im}(\omega) > 0\}$.

The susceptibility $\chi(\omega) = n^2(\omega) - 1$ obeys the Kramers-Kronig relations. In the asymptotic high-frequency limit $|\omega| \to \infty$, physical media cannot respond instantaneously to the driving field, so:

$$\lim_{|\omega|\to\infty} n(\omega) = 1. \tag{12}$$

Now consider the exponent in the Fourier integral:

$$\Phi(\omega) \equiv i\omega\left(\frac{n(\omega)x}{c_{\mathcal{I}}} - t\right). \tag{13}$$

For $|\omega| \to \infty$ in $\mathbb{C}^+$, where $\omega = \Omega + i\Xi$ ($\Xi > 0$):

$$\Phi(\omega) \to i\Omega\left(\frac{x}{c_{\mathcal{I}}} - t\right) - \Xi\left(\frac{x}{c_{\mathcal{I}}} - t\right). \tag{14}$$

If $t < x/c_{\mathcal{I}}$, then $(x/c_{\mathcal{I}} - t) > 0$. Therefore, $-\Xi(x/c_{\mathcal{I}} - t) \to -\infty$ as $\Xi \to +\infty$. The integrand decays exponentially along any semicircle of radius $R \to \infty$ in the upper half-plane. By Cauchy's residue theorem:

$$\oint_{\mathcal{C}} \tilde{E}(0,\omega)e^{\Phi(\omega)}d\omega = 0 \implies \boxed{E(x,t) \equiv 0 \quad \text{for all } t < \frac{x}{c_{\mathcal{I}}}.} \tag{15}$$

This confirms that the front velocity of information is strictly bounded by $c_{\mathcal{I}}$, and apparent superluminal group velocities ($v_g > c$) are simply pulse envelope reshaping artifacts.

3.2 Polariton Information Overhead and Quantum Memory Delay

A reciprocal question to superluminal anomalies is the subluminal regime: why does light travel slower in matter ($v < c_{\mathcal{I}}$), and what happens to the invariant speed?

While classical optics describes refraction via the Ewald-Oseen extinction theorem and secondary wave interference, at the quantum informational level deceleration is governed by **entanglement loading into matter registers**. Under Electromagnetically Induced Transparency (EIT) in a three-level Λ-atomic ensemble driven by a control field Ω_c, the propagating excitation is a **dark-state polariton** [9]:

$$\hat{\Psi}(z,t) = \cos\theta\hat{\mathcal{E}}(z,t) - \sin\theta\sqrt{N}\hat{S}(z,t), \tag{16}$$

where $\hat{S}(z,t)$ is the collective atomic spin coherence operator and $\tan\theta = g\sqrt{N}/\Omega_c$. The resulting group velocity is:

$$v_g = c_{\mathcal{I}}\cos^2\theta = \frac{c_{\mathcal{I}}}{1+\frac{g^2N}{\Omega_c^2}} = \frac{c_{\mathcal{I}}}{n_g}. \tag{17}$$

The group index n_g is revealed to be an **Information Overhead Factor**:

$$\boxed{n_g = 1 + \frac{\mathcal{I}_{\text{matter}}}{\mathcal{I}_{\text{photon}}} = 1 + \frac{g^2N}{\Omega_c^2}} \tag{18}$$

By the Mandelstam-Tamm and Margolus-Levitin Quantum Speed Limit (QSL), each transition between orthogonal atomic states requires a minimum dwell time $\tau_{\text{dwell}} \geq \pi\hbar/(2g)$. The cumulative macroscopic delay $\Delta t_{\text{delay}} = L(n_g-1)/c_{\mathcal{I}}$ is simply the sum of quantum processing dwell times inside atomic memory registers. In ultracold Bose-Einstein condensates ($v_g = 17$ m/s) [8], the excitation is 99.999994% atomic spin memory and only 0.000006% bare light, compressing a 300 m vacuum pulse to 17 μm. Spacetime's invariant speed $c_{\mathcal{I}}$ remains strictly constant.

4 Microcausality, Quantum Non-Locality, and the No-Signaling Constraint

4.1 Microcausality in Relativistic Quantum Field Theory

In relativistic Quantum Field Theory (QFT), space-like separated measurements must be causally independent. This is formalized by the **Microcausality Condition**:

$$[\hat{\mathcal{O}}_A(x), \hat{\mathcal{O}}_B(y)] = 0 \quad \text{for all } (x-y)^2 = c_{\mathcal{I}}^2(t_x-t_y)^2 - (\mathbf{x}-\mathbf{y})^2 < 0. \tag{19}$$

Let Alice measure observable $\hat{\mathcal{O}}_A$ at spacetime point x, and Bob measure $\hat{\mathcal{O}}_B$ at space-like separated point y. If Alice applies a local unitary transformation $\hat{U}_A = e^{-i\lambda\hat{\mathcal{O}}_A(x)}$ to encode a message bit, the modified expectation value at Bob's location is:

$$\langle\hat{\mathcal{O}}_B(y)\rangle' = \text{Tr}\left(\hat{U}_A\rho\hat{U}_A^\dagger\hat{\mathcal{O}}_B(y)\right). \tag{20}$$

Using the Baker-Campbell-Hausdorff expansion:

$$\hat{U}_A^\dagger\hat{\mathcal{O}}_B(y)\hat{U}_A = \hat{\mathcal{O}}_B(y) + i\lambda[\hat{\mathcal{O}}_A(x), \hat{\mathcal{O}}_B(y)] - \frac{\lambda^2}{2}[\hat{\mathcal{O}}_A(x), [\hat{\mathcal{O}}_A(x), \hat{\mathcal{O}}_B(y)]] + \dots \tag{21}$$

Since $(x-y)^2 < 0$, microcausality ensures that all commutators vanish identically:

$$[\hat{\mathcal{O}}_A(x), \hat{\mathcal{O}}_B(y)] \equiv 0 \implies \langle\hat{\mathcal{O}}_B(y)\rangle' = \langle\hat{\mathcal{O}}_B(y)\rangle. \tag{22}$$

Thus, Bob's reduced density matrix is strictly invariant:

$$\rho_B = \text{Tr}_A(\rho) \implies \frac{\partial\rho_B}{\partial\lambda} = 0 \implies \mathcal{C}_{A\to B} = 0 \quad \text{for } \Delta s^2 < 0. \tag{23}$$

4.2 Information Causality and the Tsirelson Bound

A deeper question at the interface of quantum mechanics and relativity is why nature obeys the Tsirelson bound ($S \leq 2\sqrt{2}$) rather than permitting maximal non-signaling correlations such as Popescu-Rohrlich (PR) boxes ($S = 4$). Pawłowski et al. [11] proved that this bound arises from the principle of **Information Causality**: if Alice sends an m-bit classical message to Bob at speed $v \leq c_{\mathcal{I}}$, Bob cannot access more than m bits of mutual information about Alice's database:

$$\sum_{k=0}^{N-1} I(a_k : b_k) \leq m. \tag{24}$$

Information Causality uniquely selects the Tsirelson bound $S \leq 2\sqrt{2}$ as the physical boundary of causal communication. The full complex Hilbert space structure and Born rule ($P_i = |\langle\psi_i|\psi\rangle|^2$) emerge when Information Causality is embedded within Generalized Probabilistic Theories (GPTs) satisfying operational axioms of continuous reversibility, state tomography, and purification [12]. Furthermore, in Relational Quantum Mechanics (RQM) and local unitary (Everettian/Heisenberg) formulations, state descriptions update along local future causal cones bounded by $c_{\mathcal{I}}$, eliminating the non-covariant instantaneous collapse hyperplane of textbook interpretations.

5 Emergent Spacetime from Quantum Information: The Deep Revolution

5.1 Lieb-Robinson Bounds: Emergent Light Cones in Non-Relativistic Matter

In non-relativistic quantum mechanics, the kinetic operator produces instantaneous dispersal ($c \to \infty$). However, Lieb and Robinson (1972) proved that in any quantum lattice system with short-range interactions, an effective speed limit emerges dynamically.

For any two local operators A and B supported on disjoint subsets $X, Y \subset \Lambda$ separated by distance $d(X, Y)$:

$$\|[A(t), B(0)]\| \leq C\|A\|\,\|B\| \exp\left[-\mu\left(d(X,Y) - v_{\mathrm{LR}}|t|\right)\right], \tag{25}$$

where the **Lieb-Robinson velocity** v_{LR} is given by:

$$v_{\mathrm{LR}} \leq \frac{2C\kappa}{\mu} e^1. \tag{26}$$

This shows that causal light cones are an intrinsic consequence of local quantum information processing.

5.2 Holography and the Butterfly Velocity

In holographic duality (AdS/CFT), spacetime geometry emerges from boundary entanglement. Out-of-Time-Ordered Correlators (OTOCs) characterize quantum information scrambling:

$$C(x,t) = -\langle [W(x,t), V(0)]^2 \rangle_\beta \sim \frac{1}{N^2} \exp\left[\lambda_L \left(t - \frac{|x|}{v_B}\right)\right], \tag{27}$$

where v_B is the **butterfly velocity**, which satisfies:

$$v_B = \sqrt{\frac{d}{2(d-1)}} c_{\mathcal{I}} \leq c_{\mathcal{I}}. \tag{28}$$

The event horizon of a black hole is an entanglement wedge boundary where informational reconstructibility terminates. Hence, information bounds govern spacetime geometry, not optical rays.

5.3 Entanglement Equilibrium and the Critical Audit Balance Sheet

Spacetime metric dynamics can be derived directly from the thermodynamics of quantum entanglement [13–15]. Around any event p, consider a geodesic causal diamond $\mathcal{D}(p, R)$ of radius R. The modular Hamiltonian generating conformal flow is given by:

$$\hat{K} = \frac{2\pi}{\hbar} \int_{\Sigma_0} \xi^t(r) \hat{T}_{00} d^3x, \quad \text{where } \xi^t(r) = \frac{R^2 - r^2}{2Rc_{\mathcal{I}}}. \tag{29}$$

Because $\xi^t(r)$ carries dimensions of time [s], the modular Hamiltonian $\hat{K}$ is dimensionless. Under a quantum matter excitation $T_{\mu\nu}^{\text{phys}}$, the First Law of Entanglement Entropy yields:

$$\delta S_{\text{matter}} = \delta \langle \hat{K} \rangle = \frac{2\pi}{\hbar} \int_0^R 4\pi r^2 \left(\frac{R^2 - r^2}{2Rc_{\mathcal{I}}} \right) T_{00}^{\text{phys}} dr = \frac{8\pi^2 R^4}{15\hbar c_{\mathcal{I}}} T_{00}^{\text{phys}}. \tag{30}$$

Concurrently, the Raychaudhuri equation dictates a horizon area deficit $\delta A = -\frac{4\pi R^4}{15} G_{00}^{\text{geom}}$ (with G_{00}^{geom} in units of m^{-2}), producing a boundary entanglement entropy variation:

$$\delta S_{\text{UV}} = \frac{c_{\mathcal{I}}^3}{4G\hbar} \delta A = -\frac{\pi c_{\mathcal{I}}^3 R^4}{15 G\hbar} G_{00}^{\text{geom}}. \tag{31}$$

Imposing the **Entanglement Equilibrium Hypothesis** ($\delta S_{\text{total}} = \delta S_{\text{UV}} + \delta S_{\text{matter}} = 0$) results in exact algebraic cancellation of all geometric constants:

$$-\frac{\pi c_{\mathcal{I}}^3 R^4}{15 G\hbar} G_{00}^{\text{geom}} + \frac{8\pi^2 R^4}{15\hbar c_{\mathcal{I}}} T_{00}^{\text{phys}} = 0 \implies \boxed{G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c_{\mathcal{I}}^4} T_{\mu\nu}}. \tag{32}$$

To maintain foundational clarity and methodological rigor, we explicitly establish the **Critical Audit Balance Sheet** separating derived structures from assumed empirical inputs:

Component	Status	Methodological Role
Einstein Tensor ($G_{\mu\nu}$)	**Derived**	Arises uniquely from multilinear contraction across all timelike observer 4-velocities u^μ.
Linear Coupling ($T_{\mu\nu}$)	**Derived**	Dictated by modular Hamiltonian flow generating local boost thermalization.
Coupling Scaling ($\kappa = 8\pi G/c_{\mathcal{I}}^4$)	**Derived**	Derived directly from thermodynamic bit-to-energy conversion factors.
Conservation ($\nabla^\mu T_{\mu\nu} = 0$)	**Derived**	Follows automatically from the contracted Bianchi identity $\nabla^\mu G_{\mu\nu} \equiv 0$.
Newton's Constant G	**INPUT (Assumed)**	G is not derived from pure information; it enters via the Bekenstein-Hawking coefficient $\eta = c_{\mathcal{I}}^3/4G\hbar$.
Smooth Manifold ($M, g_{\mu\nu}$)	**INPUT (Assumed)**	Assumes a pre-existing 4D smooth continuum with Riemann normal coordinates.
Entanglement Equilibrium	**INPUT (Assumed)**	Variational stationarity $\delta S_{\text{total}} = 0$ is an operational postulate analogous to the Second Law.

Table 1: Critical Audit Balance Sheet for the thermodynamic derivation of Einstein's equations.

5.4 Cosmological Horizons and Dark Energy as Information Capacity

In de Sitter spacetime bounded by an invariant causal horizon $R_{\text{dS}} = c_{\mathcal{I}}/H$, the Gibbons-Hawking entropy establishes a maximal holographic bit capacity:

$$S_{\text{dS}} = \frac{3\pi c_{\mathcal{I}}^5}{G\hbar\Lambda} \implies \Lambda = \frac{3\pi}{\ell_{\text{Pl}}^2 S_{\text{dS}}}. \tag{33}$$

While cancelling zero-point loop divergences in bulk QFT remains an open ultraviolet field-theoretic problem, the information-theoretic framework translates the 122-order-of-magnitude volume fine-tuning into an infrared boundary information capacity constraint ($S_{\mathrm{dS}} \approx 10^{122}$ bits), illustrating how a bounded speed of information $c_{\mathcal{I}}$ sets an upper bound on cosmological curvature.

6 Comparative Assessment: How Informational Causality Enhances Understanding

Dimension of Inquiry	Explanatory Gain	Foundational & Physical Understanding
1. Pedagogical & Conceptual Foundations	**Clarifies Core Intuitions**	Eradicates 120 years of fallacious paradoxes: explains why media slowing light does not alter spacetime; resolves pseudo-superluminal group velocity claims; clarifies non-signaling in quantum entanglement.
2. Axiomatic Kinematic Structure	**Structural Deepening**	Decouples spacetime geometry from electromagnetic gauge fields. Proves relativity depends on information causality, not photon masslessness ($m_\gamma = 0$).
3. Operational & Empirical Rigor	**Refined Physical Framing**	Leaves standard numerical values unchanged if $m_\gamma = 0$, while providing a precise operational framework for Proca dispersion tests and Quantum Gravity Modified Dispersion Relations (MDR).
4. Emergent Spacetime & Quantum Gravity	**Transformative Paradigm**	Reverses the ontological hierarchy: light cones are not fundamental primitives, but macroscopic consequences of microscopic Lieb-Robinson bounds and holographic scrambling velocities (v_B).

Table 2: Evaluation of how replacing the speed of light with the speed of information deepens theoretical understanding.

7 Numerical Visualizations & Analytical Parameter Models

The theoretical insights presented herein are supported by six numerical modeling suites and twenty-one publication-quality vector figures:

1. **Sommerfeld-Brillouin Wavefront Dynamics:** Numerical evaluation of Fourier-Laplace contour integrals verifying strict causal front velocity $v_{\mathrm{front}} = c_{\mathcal{I}}$ despite superluminal group delays in gain media.

2. **Proca Dispersion Phenomenology:** Evaluation of astrophysical time delays across radio to gamma frequencies, demonstrating invariant $c_{\mathcal{I}}$ amidst dispersive light.

3. **Polariton Information Overhead & Slow Light:** Analytical modeling of dark-state polariton mixing, quantum speed limit dwell time, and spatial pulse compression in matter.

4. **Lieb-Robinson Bound Visualizations:** Mapping causal operator spreading and emergent light cones in quantum many-body lattices.

5. **Quantum Gravity MDR Predictions:** Cosmological time-of-flight models for linear ($n = 1$) and quadratic ($n = 2$) Lorentz-invariance violations.

6. **Thermodynamic Horizon Dynamics:** Numerical tracking of causal diamond entropy equilibrium, black hole scrambling, and the Page curve.

8 Conclusion

The conceptual transition from the "speed of light" (c) to the "speed of information" ($c_{\mathcal{I}}$) provides a substantially clearer and more coherent understanding of fundamental physics. It clarifies relativistic kinematics, resolves century-old optical paradoxes, shields quantum causality against superluminal signaling, and correctly aligns the geometric structure of spacetime with the modern "It-from-Qubit" paradigm of emergent quantum gravity.

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